Comenius University in Bratislava
Faculty of Mathematics, Physics and Informatics

Evaluating the significance of outdoor advertising utilizing Computer Vision

Master's Thesis

2024

Bc. Carlos Andres Pizarroso Troncoso

Comenius University in Bratislava
Faculty of Mathematics, Physics and Informatics

Evaluating the significance of outdoor advertising utilizing Computer Vision

Master's Thesis

Study Programme:	Applied Computer Science
Field of Study:	Computer Science
Department:	Department of Applied Informatics
Supervisor:	RNDr. Zuzana Berger Haladová, PhD.

Bratislava, 2024

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THESIS ASSIGNMENT

Name and Surname:	Carlos Andres Pizarroso Troncoso
Study programme:	Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study:	Computer Science
Type of Thesis:	Diploma Thesis
Language of Thesis:	English
Secondary language:	Slovak

Title: Evaluating the Significance of Outdoor Advertising utilizing Computer Vision

Annotation: The goal is to create and test a method that, based on a photograph (or video), will evaluate the importance of a billboard. The work will use deep neural networks and the BillboardLamac database

Aim:
1. State of the art review of deep learning object detectors and trackers
2. Choose 3 methods for object detection on image/videos and test them on the Billboard Lamac dataset
3. Create a new method for classification of billboard significance
4. Validate the propose method

Literature:
Zou, Zhengxia, et al. "Object detection in 20 years: A survey." Proceedings of the IEEE (2023).
H. Marciano et al., “The effect of billboard design specifications on driving: a pilot study,” Accident Analysis & Prevention, vol. 104, pp. 174–184, 2017.
Z. Bylinskii, T. Judd, A. Oliva, A. Torralba, and F. Durand, “What do different evaluation metrics tell us about saliency models?” IEEE transactions on pattern analysis and machine intelligence, vol. 41, no. 3, pp. 740–757, 2018.

Supervisor:	RNDr. Zuzana Berger Haladová, PhD.
Department:	FMFI.KAI - Department of Applied Informatics
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Assigned:	06.10.2023
Approved:	14.11.2023

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Guarantor of Study Programme

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ZADANIE ZÁVEREČNEJ PRÁCE

Meno a priezvisko študenta: Carlos Andres Pizarroso Troncoso
Študijný program: aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor: informatika
Typ záverečnej práce: diplomová
Jazyk záverečnej práce: anglický
Sekundárny jazyk: slovenský

Názov: Evaluating the Significance of Outdoor Advertising utilizing Computer Vision
Vyhodnocovanie dôležitosti billboardov z využitím počítačového videnia

Anotácia: Cieľom je vytvoriť a otestovať metódu, ktorá na základe fotografie (alebo videa) vyhodnotí dôležitosť billboardu. Práca bude využívať hlboké neurónové siete a databázu BillboardLamac.

Cieľ:
1. State of the art review of deep learning object detectors and trackers
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Literatúra:
Zou, Zhengxia, et al. "Object detection in 20 years: A survey." Proceedings of the IEEE (2023).
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Z. Bylinskii, T. Judd, A. Oliva, A. Torralba, and F. Durand, “What do different evaluation metrics tell us about saliency models?” IEEE transactions on pattern analysis and machine intelligence, vol. 41, no. 3, pp. 740–757, 2018.

Vedúci: RNDr. Zuzana Berger Haladová, PhD.
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Vedúci katedry: doc. RNDr. Tatiana Jajcayová, PhD.

Dátum zadania: 06.10.2023

Dátum schválenia: 14.11.2023

prof. RNDr. Roman Ďurikovič, PhD.
garant študijného programu

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študent

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vedúci práce

Acknowledgments:

Abstrakt

Kľúčové slová: Keywords

Abstract

Keywords: Keywords

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Chapter 1

Terminology

An example chapter.

Chapter 2

Pipeline

An example chapter with sections.

2.1 Object Detector

An example section.

2.2 Classifier

An example section.

Chapter 3

Results

An example chapter.

Conclusion

My conclusion.

Bibliography

[1] David Crundall, Editha Van Loon, and Geoffrey Underwood. Attraction and distraction of attention with roadside advertisements. *Accident Analysis and Prevention*, 38(4):671–677, July 2006.

[2] Zuzana Černeková, Zuzana Berger Haladová, Ján Špirka, and Viktor Kocur. Evaluating the significance of outdoor advertising from driver's perspective using computer vision. *2023 World Symposium on Digital Intelligence for Systems and Machines (DISA)*, pages 53–58, November 2023.

Appendix A

Myproject source code

Appendix B

Benchmarking results